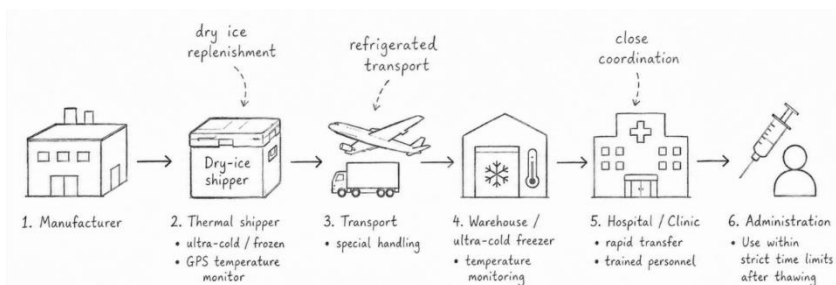


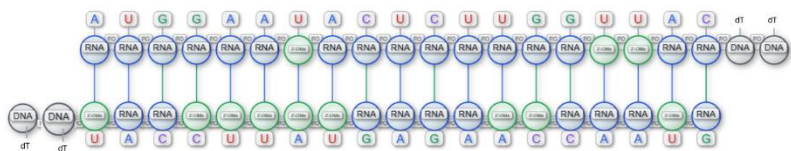
mRNA vaccines are highly temperature-sensitive because they are vulnerable to degradation through mRNA hydrolysis, lipid oxidation, RNA–lipid chemical interactions and nanoparticle aggregation, each of which can reduce potency during storage and distribution. During the pandemic, Pfizer-BioNTech’s mRNA vaccine initially required ultra-cold storage at approximately -70°C , whereas Moderna’s vaccine required frozen storage at around -20°C . These requirements created substantial logistical burdens across packaging, transportation, warehousing and administration (Fahrni et al., 2022).

For example, Pfizer-BioNTech’s vaccine had to be shipped in dry-ice containers with GPS-enabled temperature monitors, transferred rapidly to ultra-cold freezers after arrival and used within strict time limits after thawing. If storage was prolonged or disrupted, dry ice had to be replenished to maintain appropriate conditions (Fahrni et al., 2022). Such constraints required specialized freezers, validated thermal shippers, temperature-monitoring systems, trained personnel, refrigerated transport and close coordination across manufacturers, freight providers, airports, warehouses and healthcare facilities (Fahrni et al., 2022).



The COVID-19 vaccine experience therefore highlighted cold-chain dependence as both a formulation problem and an access problem. Improving RNA-LNP stability through formulation optimization, lyophilization, improved packaging and reduced temperature sensitivity is essential not only for preserving therapeutic efficacy, but also for lowering distribution costs, reducing wastage and improving equitable global access.

Example siRNA/LNP



Patisiran (Onpattro), an LNP-formulated RNAi therapeutic, established that chronic systemic delivery of siRNA is clinically feasible. However, its safety and administration burden reflect the inherent liabilities of lipid nanoparticles. In the APOLLO trial, patients treated with patisiran experienced higher rates of infusion-related reactions (19% vs 9% with placebo) and peripheral edema (30% vs 22%), along with systemic adverse events such as fatigue, diarrhea, and dizziness. These effects are consistent with known LNP-associated mechanisms, including complement activation, innate immune stimulation, and vascular responses. As a result, patisiran requires intravenous infusion and routine premedication (e.g., corticosteroids, antihistamines) to mitigate hypersensitivity and infusion reactions.

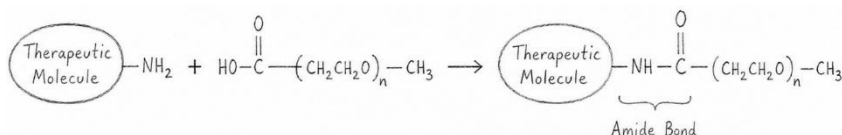
In contrast, vutrisiran is a next-generation RNAi therapeutic targeting the same gene (TTR) but uses a GalNAc conjugate rather than LNPs. This enables direct hepatocyte uptake via the asialoglycoprotein receptor, avoiding many of the systemic immune interactions associated with nanoparticle carriers. Consequently, vutrisiran can be administered as a subcutaneous injection once every three months, substantially reducing treatment burden.

Patisiran (Onpattro)	Category	Vutrisiran
TTR RNAi therapeutic	Target	TTR RNAi therapeutic
LNP-formulated siRNA	Delivery platform	GalNAc-conjugated siRNA
Systemic nanoparticle delivery	Hepatocyte uptake	Direct ASGPR-mediated uptake
Intravenous infusion	Administration	Subcutaneous injection
Repeat IV treatment visits	Dosing schedule	Once every 3 months
Required	Premedication	Not required
Infusion-related reactions observed Peripheral edema more common	Immune / infusion signal	No infusion-related reaction signal
Infusion reactions, peripheral edema, fatigue, diarrhea, dizziness	Common tolerability profile	Pain in extremity, arthralgia, dyspnea, decreased vitamin A
Higher: infusion center + monitoring + premedication	Treatment burden	Lower: simple repeat dosing paradigm

Vutrisiran demonstrates a distinct and generally improved tolerability profile. In the HELIOS-A study, the most common adverse events in patients with hATTR-PN were pain in extremity (15%), arthralgia (11%), dyspnea (7%), and decreased vitamin A levels (7%), with no signal of infusion-related reactions. Similarly, in ATTR-CM, no new safety concerns were identified. Notably, unlike patisiran, vutrisiran does not require premedication, underscoring the advantage of eliminating LNP-associated immune activation and enabling a more convenient, repeat dosing paradigm.

ONPATTRO requires weight-based IV infusion every 3 weeks, along with premedication (corticosteroids, acetaminophen, and antihistamines) and monitoring due to infusion-related reactions and broader systemic exposure. In contrast, GalNAc-conjugated siRNAs achieve targeted hepatocyte uptake via the asialoglycoprotein receptor, enabling subcutaneous administration, fixed dosing, and much less frequent schedules (monthly to every 3–6 months or longer) without routine pre-treatment. Overall, GalNAc conjugation has enabled more precise liver targeting and more convenient, less intrusive dosing compared with earlier non-targeted LNP approaches like ONPATTRO.

PEGylation



PEGylation, the covalent attachment of polyethylene glycol (PEG) to therapeutic molecules. The technology was discovered in the 1960s by Frank Davis who proposed its use to reduce the immunogenicity of protein drugs by shielding them from immune recognition and prolonging circulation time (Veronese & Pasut, 2005; Pasut & Veronese, 2012). This concept was experimentally validated in a landmark study by Abuchowski et al. (1977), which demonstrated that PEGylation of bovine liver catalase significantly decreased immunogenicity while extending plasma half-life.

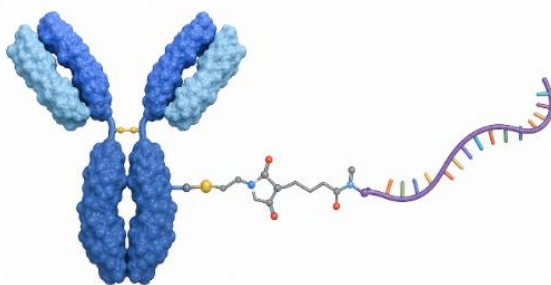
PEGylation transitioned from academic concept to commercial technology (Harris & Chess, 2003) with the FDA approval of Adagen in 1990, followed by Oncaspar in 1994. (Alconcel et al., 2011; Harris & Chess, 2003). From the mid-1990s onward, PEGylation rapidly expanded across diverse drug classes and architectures, including the development of PEGylated liposomes such as Doxil (Allen & Cullis, 2013), site-specific

mono-PEGylated proteins like Pegintron, and branched PEG systems exemplified by Pegasys (Veronese & Mero, 2008).

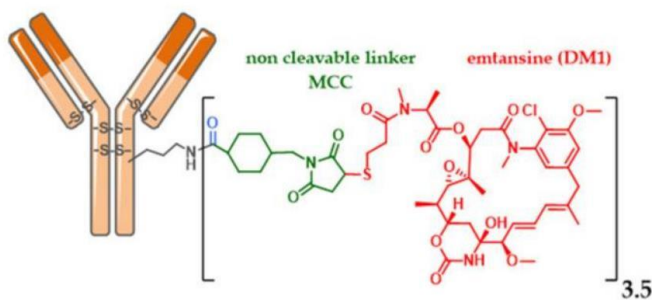
The technology further diversified to include PEGylated aptamers (Macugen™), antibody fragments (Cimzia™), peptides (Omontys), and small molecules (Movantik), reflecting its broad applicability across therapeutic modalities (Keefe et al., 2010; Fishbane et al., 2013). More recently, PEGylation has played a critical role in nanomedicine, particularly in lipid nanoparticle (LNP) systems used for nucleic acid delivery; a critical component of the Onpattro LNP and the widespread deployment of PEGylated LNP-based mRNA COVID-19 vaccines (Cullis & Hope, 2017; Hou et al., 2021).

While there are many more ligand conjugate technologies used including palmitate and antibodies most pro-drug technologies for tissue specificity is for liver with various forms of GalNAc.

Antibody conjugation



Antibody–oligonucleotide conjugates (AOCs) are an emerging delivery platform that couples the tissue specificity of monoclonal antibodies with the gene-modulating activity of oligonucleotide therapeutics, including siRNA, antisense oligonucleotides (ASOs), and related chemistries. In these constructs, oligonucleotides are covalently attached to antibodies or antibody fragments through linkers and bioconjugation strategies such as maleimide, disulfide, lysine–amine, or cysteine–thiol coupling, with the goal of promoting receptor-mediated uptake and intracellular delivery to otherwise difficult-to-access tissues (Junutula et al., 2008; Cuellar et al., 2015). Early antibody–siRNA conjugates demonstrated that chemically defined antibody conjugates could retain both antibody binding and RNA-silencing activity, although efficient



anti-HER2 Kadcyla® (ado-trastuzumab emtansine)

Kadcyla®

intracellular trafficking and endosomal escape remained important barriers to broader therapeutic translation (Cuellar et al., 2015).

Antibody-Drug Conjugates (ADC) began with first-generation constructs that used murine, chimeric, or early humanized antibodies, relatively unstable linkers, and low-potency cytotoxic payloads such as doxorubicin or duocarmycins. These agents established the feasibility of antigen-directed delivery but were limited by heterogeneous lysine conjugation, broad drug-to-antibody ratio distributions, premature payload release, immunogenicity, and narrow therapeutic indices, as seen with early approved agents such as Mylotarg® and Besponsa® (Beck et al., 2017; Tsuchikama and An, 2018; Shastry et al., 2023).

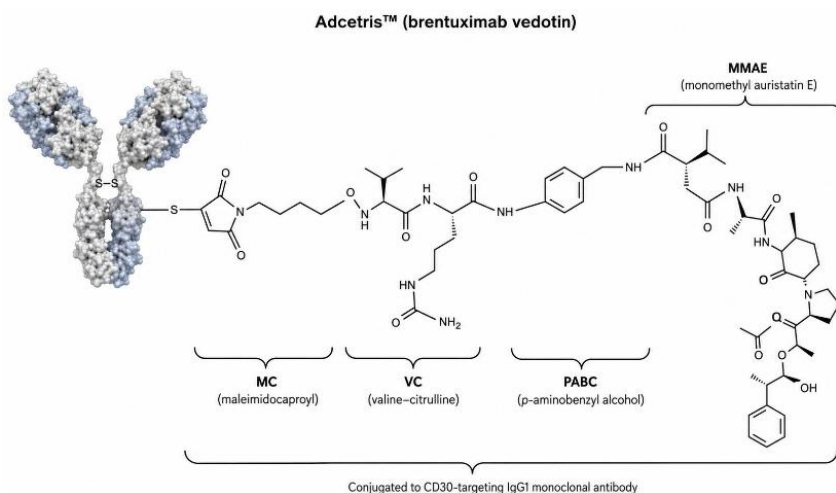
Second-generation ADCs improved this framework through more humanized antibodies, more stable linkers, and higher-potency payloads, particularly auristatins and maytansinoids. Compounds such as Adcetris®, Kadcyla®, Padcev®, and Elahere® showed that ADCs could produce meaningful clinical benefit across hematologic malignancies and solid tumors, but random lysine or reduced interchain cysteine conjugation continued to generate heterogeneous products, and high-DAR species could be associated with rapid clearance, off-target toxicity, or acquired resistance (Beck et al., 2017; Drago et al., 2021; Shastry et al., 2023).

Third-generation ADCs increasingly emphasize molecular precision: fully human antibodies or Fab-based formats, stable circulation-compatible linkers, site-specific conjugation, and defined DAR values. This engineering logic underlies newer agents such as Enhertu® and Trodelvy®, which use optimized linker–payload systems and topoisomerase I inhibitor payloads to expand activity in tumors with low or heterogeneous antigen expression, in part through bystander killing (Drago et al., 2021; Shastry et al., 2023).

Like modern ADCs, successful AOCs will likely depend on precise control of antibody format, conjugation site, linker stability, payload

loading, and intracellular trafficking. However, AOCs face additional challenges imposed by oligonucleotide size, charge, nuclease sensitivity, endosomal escape, and the need to deliver the payload to the cytosol or nucleus rather than simply to the lysosome. Maturation of AOCs will similarly require iterative improvements in linker chemistry, site-specific conjugation, payload design, and a deeper understanding of tissue-specific uptake and intracellular processing.

Site-specific platforms, including Genentech's THIOMAB technology, improve conjugate homogeneity and pharmacokinetic performance by introducing defined cysteine attachment sites on antibodies (Nanna et al., 2020). AOCs have shown strong potential for extrahepatic delivery, especially to muscle and central nervous system tissues, where conventional oligonucleotide delivery remains difficult. For example, anti-CD71 antibody–siRNA conjugates have achieved efficient gene silencing in skeletal muscle and cardiac tissue, and companies such as Denali Therapeutics and Avidity Biosciences have advanced transferrin receptor–targeting antibody–oligonucleotide systems linked through noncleavable maleimide chemistry (Sugo et al., 2016). By exploiting receptor-mediated internalization, these approaches can improve tissue targeting, intracellular delivery, and the overall therapeutic index, positioning AOCs as a promising next-generation platform for precision RNA therapeutics (Winkler, 2013; Belgrad, Fakih and Khvorova, 2024).



Adcetris™ (brentuximab vedotin) provides a useful template for developing a new antibody–oligonucleotide conjugate (ADO/AOC) with a cleavable linker. In Adcetris, a CD30-targeting IgG1 antibody is joined to the cytotoxic payload MMAE through the MC–VC–PABC linker. The MC portion, maleimidocaproyl, serves as the antibody-attachment handle; its